

## Assessment Tool - 3

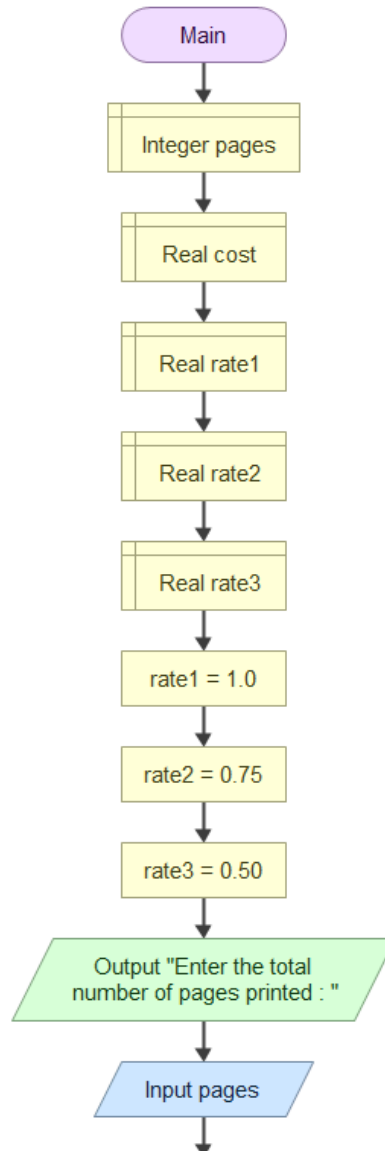
### Writing Pseudo Code and its Flow Chart

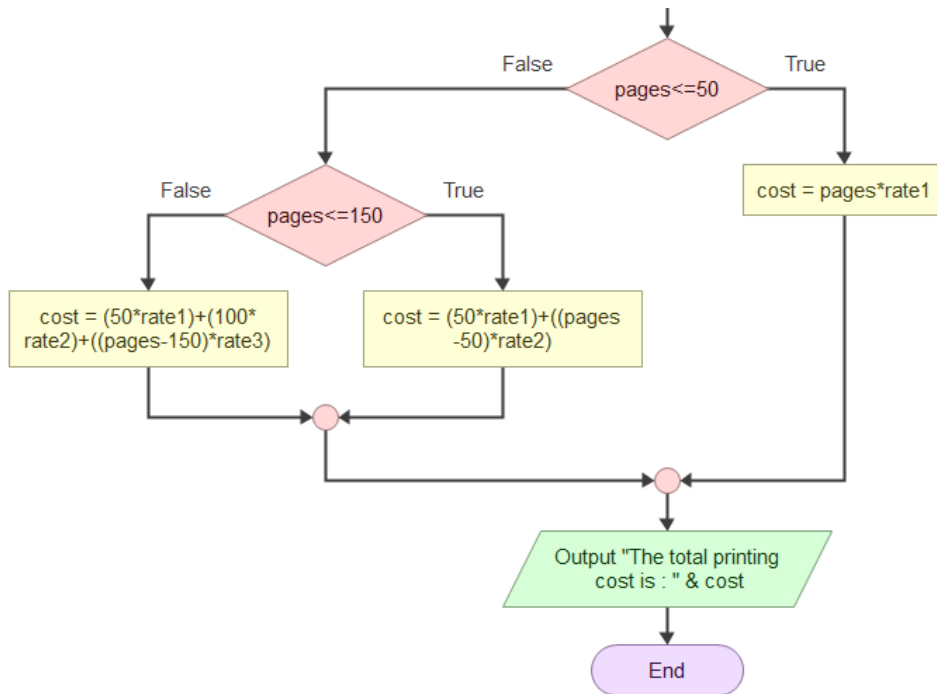
**Name :** Tamil Elakiya S

**Reg No :** 192524443

**1 . Read the total number of pages printed and calculate printing cost with separate rates for the first 50, next 100, and remaining pages.**

**Flowchart :**





### Pseudo Code :

Function Main

Declare Integer pages

Declare Real cost

Declare Real rate1

Declare Real rate2

Declare Real rate3

Assign rate1 = 1.0

Assign rate2 = 0.75

Assign rate3 = 0.50

Output "Enter the total number of pages printed : "

Input pages

If pages <= 50

    Assign cost = pages \* rate1

Else

    If pages <= 150

```

        Assign cost = (50*rate1)+((pages-50)*rate2)
    Else
        Assign cost = (50*rate1)+(100*rate2)+((pages-150)*rate3)
    End
End
Output "The total printing cost is : " & cost
End

```

### **Java Code :**

```

import java.util.*;
import java.lang.Math;

public class Program {
    private static Scanner input = new Scanner(System.in);

    public static void main(String[] args) {
        int pages;
        double cost;
        double rate1;
        double rate2;
        double rate3;

        rate1 = 1.0;
        rate2 = 0.75;
        rate3 = 0.5;
        System.out.println("Enter the total number of pages printed : ");
        pages = input.nextInt();
        if (pages <= 50) {
            cost = pages * rate1;
        } else {
            if (pages <= 150) {

```

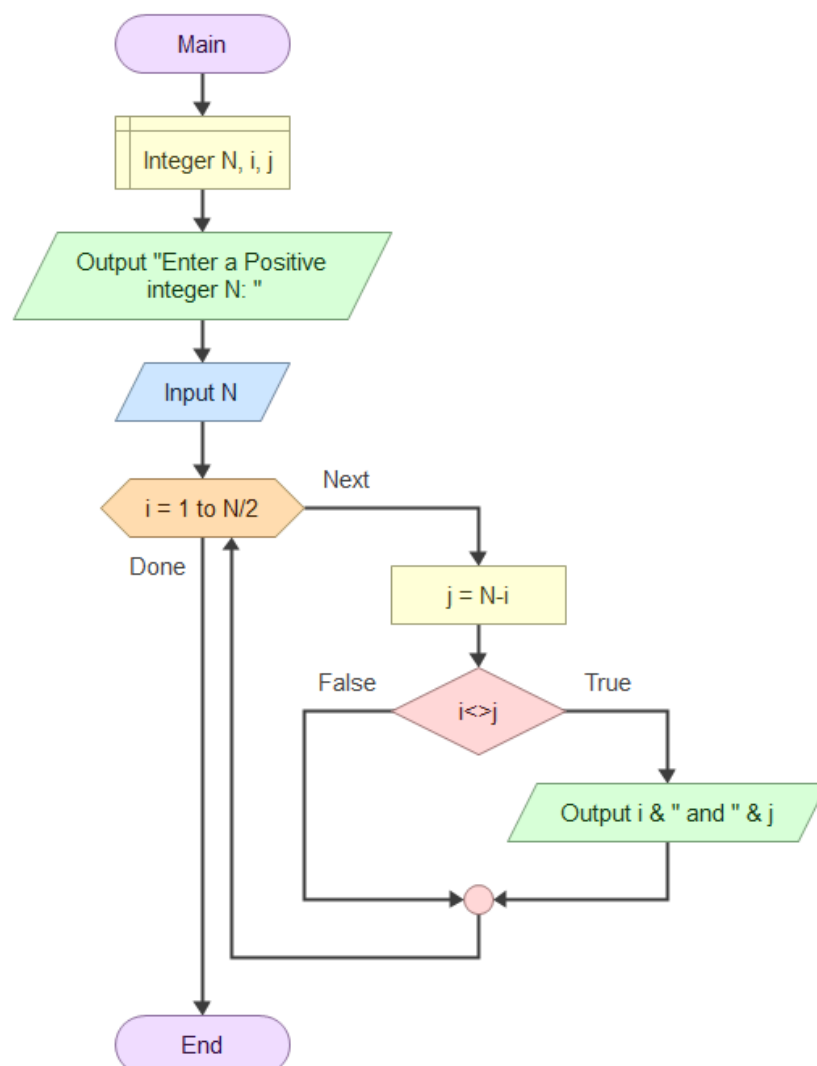
```

        cost = 50 * rate1 + pages - 50 * rate2;
    } else {
        cost = 50 * rate1 + 100 * rate2 + pages - 150 * rate3;
    }
}
System.out.println("The total printing cost is : " + cost);
}
}

```

**2 . Read an integer N and print all pairs of positive integers whose sum equals N.**

**Flowchart :**



### **Pseudo Code :**

```
Function Main
  Declare Integer N, i, j

  Output "Enter a Positive integer N: "
  Input N
  For i = 1 to N/2
    Assign j = N-i
    If i<>j
      Output i & " and " & j
    End
  End
End
End
```

### **Java Code :**

```
import java.util.*;
import java.lang.Math;

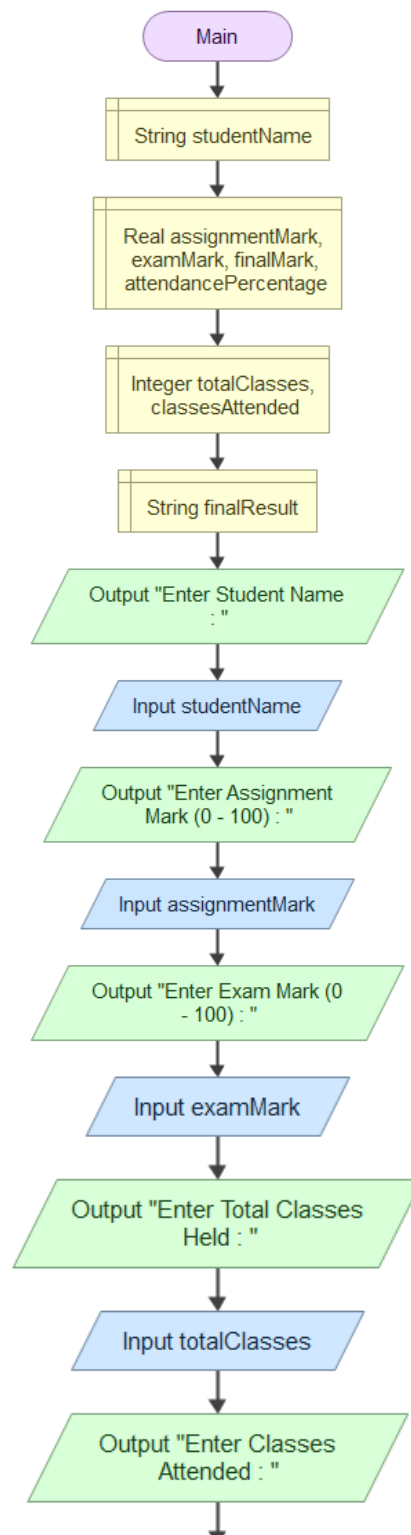
public class Program {
  private static Scanner input = new Scanner(System.in);

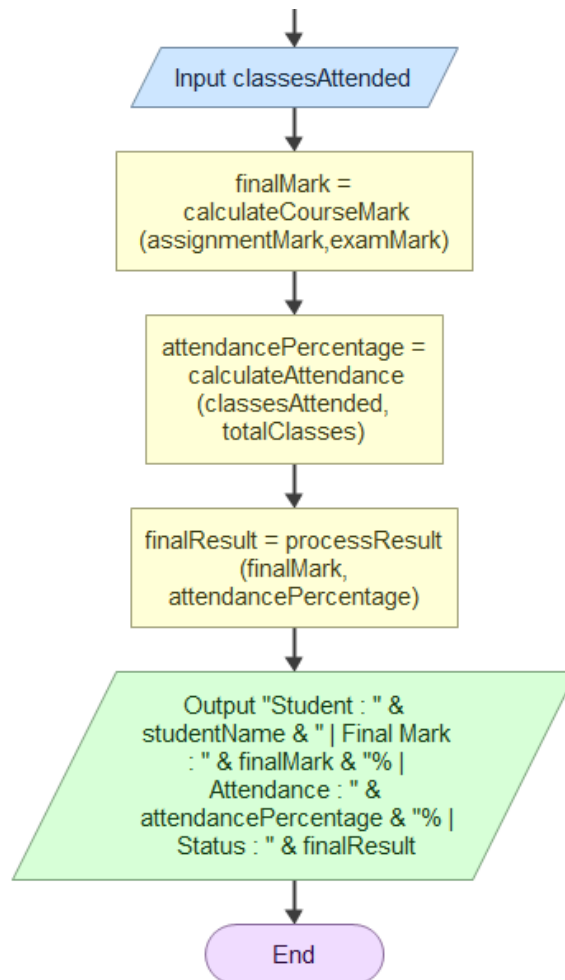
  public static void main(String[] args) {
    int n, i, j;

    System.out.println("Enter a Positive integer N: ");
    n = input.nextInt();
    for (i = 1; i <= (double) n / 2; i++) {
      j = n - i;
      if (i != j) {
        System.out.println(Integer.toString(i) + " and " + j);
      }
    }
  }
}
```

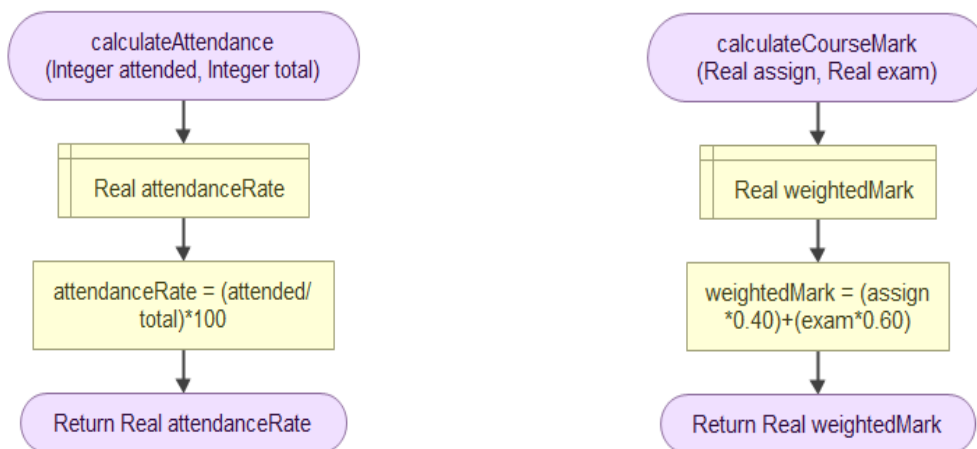
3 . Design an academic-report flowchart with CourseMark, Attendance, and Result processing functions. Explain how these could be organised into separate Java packages.

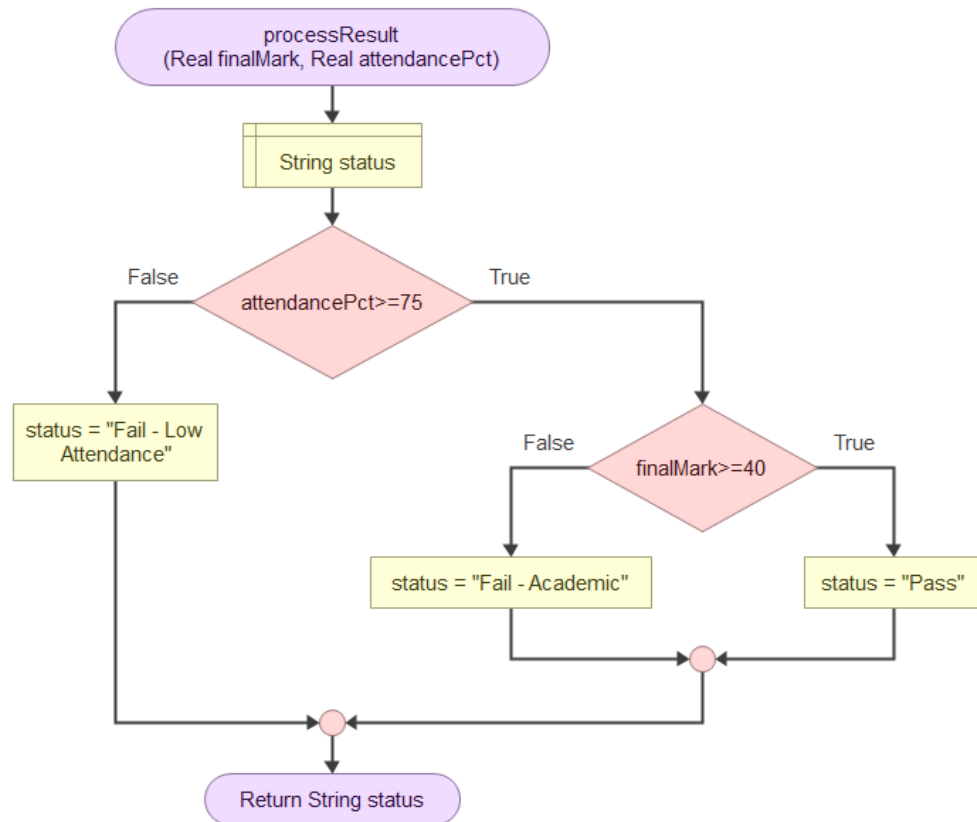
Flowchart :





### Functions :





### **Pseudo Code :**

Function Main

Declare String studentName

Declare Real assignmentMark, examMark, finalMark, attendancePercentage

Declare Integer totalClasses, classesAttended

Declare String finalResult

Output "Enter Student Name : "

Input studentName

Output "Enter Assignment Mark (0 - 100) : "

Input assignmentMark

Output "Enter Exam Mark (0 - 100) : "

Input examMark

Output "Enter Total Classes Held : "

Input totalClasses



```
Output "Enter Classes Attended : "  
Input classesAttended  
Assign finalMark = calculateCourseMark(assignmentMark,examMark)  
Assign attendancePercentage =  
calculateAttendance(classesAttended,totalClasses)  
Assign finalResult = processResult(finalMark,attendancePercentage)  
Output "Student : " & studentName & " | Final Mark : " & finalMark & "% |  
Attendance : " & attendancePercentage & "% | Status : " & finalResult  
End
```

```
Function calculateAttendance (Integer attended, Integer total)
```

```
    Declare Real attendanceRate
```

```
    Assign attendanceRate = (attended/total)*100
```

```
    Return Real attendanceRate
```

```
End
```

```
Function calculateCourseMark (Real assign, Real exam)
```

```
    Declare Real weightedMark
```

```
    Assign weightedMark = (assign*0.40)+(exam*0.60)
```

```
    Return Real weightedMark
```

```
End
```

```
Function processResult (Real finalMark, Real attendancePct)
```

```
    Declare String status
```

```
    If attendancePct>=75
```

```
        If finalMark>=40
```

```
            Assign status = "Pass"
```

```
        Else
```

```
        Assign status = "Fail - Academic"
    End
Else
    Assign status = "Fail - Low Attendance"
End
Return String status
End
```

### **Java Code :**

```
import java.util.*;
import java.lang.Math;

public class Program {
    private static Scanner input = new Scanner(System.in);

    public static void main(String[] args) {
        String studentName;
        double assignmentMark, examMark, finalMark, attendancePercentage;
        int totalClasses, classesAttended;
        String finalResult;

        System.out.println("Enter Student Name : ");
        studentName = input.nextLine();
        System.out.println("Enter Assignment Mark (0 - 100) : ");
        assignmentMark = input.nextDouble();
        System.out.println("Enter Exam Mark (0 - 100) : ");
        examMark = input.nextDouble();
        System.out.println("Enter Total Classes Held : ");
        totalClasses = input.nextInt();
        System.out.println("Enter Classes Attended : ");
        classesAttended = input.nextInt();
```

```
        finalMark = calculateCourseMark(assignmentMark, examMark);
        attendancePercentage = calculateAttendance(classesAttended, totalClasses);
        finalResult = processResult(finalMark, attendancePercentage);
        System.out.println("Student : " + studentName + " | Final Mark : " + finalMark
+ "% | Attendance : " + attendancePercentage + "% | Status : " + finalResult);
    }
```

```
public static double calculateAttendance(int attended, int total) {
    double attendanceRate;

    attendanceRate = (double) attended / total * 100;

    return attendanceRate;
}
```

```
public static double calculateCourseMark(double assign, double exam) {
    double weightedMark;

    weightedMark = assign * 0.4 + exam * 0.6;

    return weightedMark;
}
```

```
public static String processResult(double finalMark, double attendancePct) {
    String status;

    if (attendancePct >= 75) {
        if (finalMark >= 40) {
            status = "Pass";
        } else {
            status = "Fail - Academic";
        }
    }
}
```

```
    }  
  } else {  
    status = "Fail - Low Attendance";  
  }  
  
  return status;  
}  
}
```